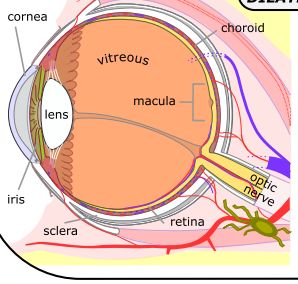
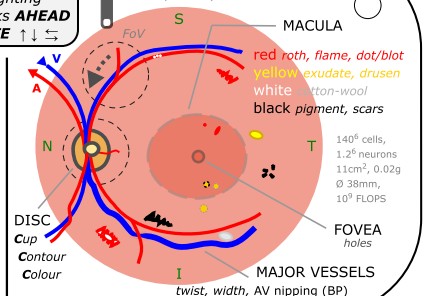
15 ANATOMY

> Get **CLOSE DIM** lighting
> Px looks **AHEAD DILATE** ↑↓ ⇆

arm's-length - reflex
hands-span +4 - details



"Look for defects everywhere especially at the macula"

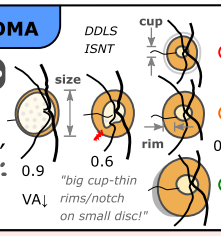


11 GLAUCOMA

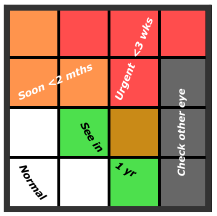
OPEN Angle slow periph field↓
1⁰ or 2⁰ uveitis, neovasc, pigment, steroid, exfoliation

> **Discs** R<>L, **IOP, VA, Fields, Pupils** 12
"powerful together"
> **Drops** ongoing **LASER, Surgery**

DDLS ISNT



C/D Ratio Small +0.2 … Large -0.2
0-0.2 0.3-0.5 0.6-0.8 0.9-1



IOP *true*
≥30
25-29
20-24
≤20

CLOSED Angle pain+++, VA↓/haloes, nausea
risks: *age, ♀, race, cataract, hyperopic*
mid-dilated unreactive pupil, red, iris bow, hazy cornea, IOP↑↑ ~40+
> **Diamox/Pilocarpine/Steroids massage > LASER, IOL**

risks: *discs, IOP, age, race, family, myopia, diabetic, CCT↓*

8 VISION DEVELOPMENT

Birth - 3mth	3- 6mth	6-12 mth	12-18 mth	18mth - 8yr
Bright light: *squeezes*	3D vision (no squint)	Familar v Stranger	Toy hide: *searches*	Adult-like yet plastic
Light on/off: *still/widen*	Reach: *rough*	Reach: *accurate*	Pictures: *point to detail*	Mature by 8yr
Sees black/white/bright	Smooth follow	Interest in books	Vision + walk: *clumsy*	
Squints: *sometimes*	Returns silent smile			
Fix: *only on one*	Fix: *switches*			

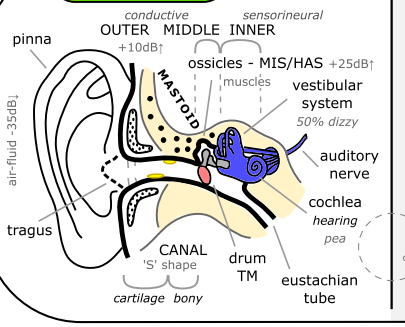
Fix + Follow Faces

Age	Dist (m)	VA
1wk	0.30	6/600
2wk	0.75	6/180
6wk	1.15	6/90
3mth	1.50	6/60
6mth	2.00	6/36
18mth	3.00	6/18

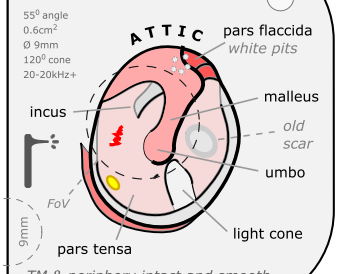
Childhood VI

No smile return	Fast wobbly eyes	Stare at lights	Unhappy busy
Not reaching	Roving eyes	Finger flicking	Clutter hard
Startled easily		Eye poking	
Hold close			

8 ANATOMY



"Look for defects everywhere especially in the attic"



TM & periphery intact and smooth

20 REFRACT

	VA	Pinhole	Near	
Normal	✓		✓	
Myopia (short-sight)	↓	↑	✓	> *MINUS (minimum)*
Hyperopia (long-sight)	↓	↑	↓	> *PLUS lenses (maximum)*
Presbyopia (age)	✓		↓	> *PLUS lenses (working dist)*
Not sure	↓	X	↓	> *Cataract/pathology*

Glasses > *ear+nose fit, tilt, bow, centration*

Age						aphakic +10D
40	50	60	70	80	IOL	
+0.75	+1.50	+2.00	+2.50	+2.75	+3.00	

5 CHILD

Congenital Profound loss
1:1000 mostly sensorineural, genetic, low birth weight, early infection, poor nutrition
> *counselling, immunisation, education & hygiene, SURGERY (cochlear), social help - **6mth***

> **full sign language**

if baby sleeps well = no urgent ear probs

MENINGITIS ! headaches, vomiting, fever++, drowsy, stiff neck

Co-ordination

Hearing Development
0-8wk - sounds startle, opens eyes, blinks, looks alert, enjoys singing
6mth - turns head or eyes in direction of sound, likes being talked to
9mth - listens to loud and soft sounds, makes all sorts of babbling noises
12mth - responds to own name, starting 'baby' words
18mth - points to familar objects when asked, using several simple words
2yr - talks and puts words together, responds to even soft voices DEAFBLIND rubella

observation, distraction, matching

Speech/Lang Development
PATIENCE
> *get close/good light look straight ahead remove distractions repeat words over encourage lip reading don't cover face or eat nor 'overprotect'*

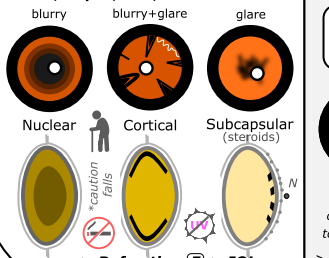
WITHDRAWN/'STUPID'
> *core tests* 6
home signing engage teacher hearing aids

"Healthy Ear" game days

7 LENS

projection+RAPD

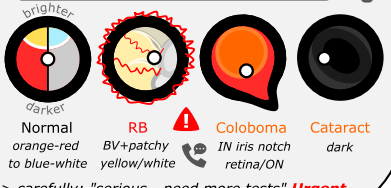
> **VA, Pupils, IOP, Dilate**



> **Refraction** 5 **> IOL**

Fundal+Corneal Reflex know the glow
childhood: cataract 1:1000,
coloboma 1:5000, retinoblastoma (RB) 1:15,000+

> *dim room+arm's-length compare eyes* ***light, shape, crescent, colour, angle*** *birth, 6wk, routine visits, any concerns* 13



Normal	RB	Coloboma	Cataract
orange-red to blue-white	*BV+patchy yellow/white*	*IN iris notch retina/ON*	*dark*

> *carefully:* "serious - need more tests" **Urgent**

2 EAR

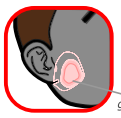
AROUND EAR
> **LOOK+FEEL** *from all angles; check TMJ, tenderness, neuralgia*

Swollen+Sore
Parotitis: mumps, bacterial



1 or 2 sides

Firm/nodular mass
Tumour



Swollen+Pain
Mastoiditis



> *ABx*
Incise

look in ear too

PINNA
size & shape, fistulas, scars

Sore/Itchy
> *antiseptic cream*



Eczema-watery
Trauma-CSF

Swollen+Red
Cellulitis > *oral ABx*



Swollen+Pain
Pre-Auricular Sinus

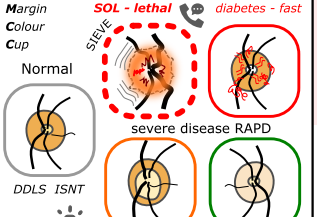


> *ABx*
Incise

8 FUNDUS 1

The 5 Discs
Margin
Colour
Cup

SWOLLEN ! *SOL - lethal*
New Vessels *diabetes - fast*



Diabetic Retinopathy
inflam 2 heart, brain, kidneys, nerves, feet, eyes
> ***SCREEN: BIO or DO***

race
5yr
Non-Proliferative (*Watch*)
Proliferative (*Urgent*)

RETINOPATHY
MACULOPATHY
'FOVEOPATHY'

MA, D/B, CWS, VB
Macula HE
> bp, lipids HbA1c
Fovea HE + odem = VA↓
NVD, Vit H, PR-H, NVE
N0↑

Sleep/Eat/Move/Think/Sun > *aVEGF > LASER*

seconds
ARTERIAL Occlusion <3/60, embolic, GCA
central+branch: 6hrs, pale red spot > *BIO R<>L*
> ***BP, DM, ESR, lipids, USS - stroke risk***

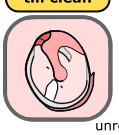
minutes
VEIN Occlusion swollen disc+mac, tort bv's
central+branch: wake up, "blood & thunder"
isc: <3/60, RAPD - NVD/E/I, vit H > ***PRP***
non-isc: >6/60, wait

4 DRUM

> *hold like a pen* ***TILT*** *head, pull pinna* ***STRAIGHTEN*** *canal* ***AIM*** *at back of eye check 'good' ear first*

Systematically check Drum+ATTIC

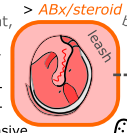
dry mop till clean



cold, throat, pain, fever

unresponsive > ***Myringotomy***

AOM-Red+Sore
> *ABx/steroid*



CSOM-Wet Hole
> *mop+ABx drops*



chronic 6wk+ painless!

"Sore Ear"

heals in 6mth

Dry > *test*
Hole *hearing, future*

retraction - *'sucked'*
holes - *old injury*
scars - *'chalk' patch*
white pits - *disease*
blood - *trauma*
inflam - *bulge*
mobile

'Angry' mass+smell
Cholesteatoma



OME-Dull/Sucked (Glue)
> *wait, clears 3mth*

